

D. Cross-Motif Regulatory Avoidance Summary

Universally Avoided

Regulatory/TF (0.60x) *

Hypothetical (0.53x) *

Universally Enriched

DNA/RNA metabolism (1.59x) *

Stress/Defense (1.60x)

Membrane/Cell wall (1.38x)

Motif-Specific / No Pattern

Primary metabolism

Transport

Secondary metabolism

Other

Translation

* = Bonferroni $p < 0.05$ in at least one motif group

Analysis based on 8 motif-timepoint groups, 7996 protein-coding genes, 2kb proximity window

VERDICT

AAGCCCG 6mA T1: Regulatory/TF fold=0.43, p_bonf=0.017

GCCGGC 4mC T1: Regulatory/TF fold=0.61, p_bonf=0.000

UNIVERSAL pattern